Water balance

Purpose

Water balance files report the major flows and storages of the HRU-to-channel hydrology: precipitation, snow, ET, surface runoff, lateral and tile flow, percolation, soil-water storage, and snow pack. One file is written per scope (`hru`, `hru-lte`, `lsunit`, `ru`, `region`, `basin`) per interval (`day`, `mon`, `yr`, `aa`), toggled in `print.prt`.

Source

- HRU writer: src/hru_output.f90, units 2000 (day), 2001 (mon), 2002 (yr), 2003 (aa).
- Landscape unit writer: src/lsu_output.f90.
- Routing unit writer: src/ru_output.f90.
- Basin writer: src/basin_output.f90.
- Region writer: src/lsreg_output.f90.
- File opening and header writes: src/output_landscape_init.f90.
- Record type: src/output_landscape_module.f90, `type output_waterbal`.

Format

- Line 1: title (project name, version banner).
- Line 2: column header (`wb_hdr`).
- Line 3: units (`wb_hdr_units`).
- Lines 4+: data records. One record per object (HRU, lsunit, …) per timestep.

The leading seven columns are the same for every landscape output:

#	Column	Type	Description
1	jday	int	Julian day of year
2	mon	int	calendar month
3	day	int	day of month
4	yr	int	calendar year
5	unit	int	spatial id (HRU number, lsunit number, …)
6	gis_id	int	GIS id from the object table
7	name	char	object name (`ob(iob)%name`)

The remaining columns come from `output_waterbal`, in declaration order. All flows are mm of water depth over the unit area; CN is dimensionless.

#	Column	Units	Description
8	precip	mm	precipitation (rain + snow)
9	snofall	mm	precipitation falling as snow, sleet, or freezing rain
10	snomlt	mm	snow or melting ice
11	surq_gen	mm	surface runoff generated on the landscape
12	latq	mm	lateral soil flow
13	wateryld	mm	water yield (surq + latq + tile)
14	perc	mm	percolation out of soil profile into vadose zone
15	et	mm	actual evapotranspiration

#	Column	Units	Description
16	ecanopy	mm	canopy evaporation (not reported in this column in current source)
17	eplant	mm	plant transpiration
18	esoil	mm	soil evaporation
19	surq_cont	mm	surface runoff leaving the landscape
20	cn	none	average curve number for the timestep
21	sw_init	mm	soil water content at start of timestep
22	sw_final	mm	soil water content at end of timestep
23	sw_ave	mm	average soil water content over the timestep
24	sw_300	mm	final soil water content in upper 300 mm
25	sno_init	mm	snow water equivalent at start of timestep
26	sno_final	mm	snow water equivalent at end of timestep
27	snopack	mm	average snow water equivalent over the timestep
28	pet	mm	potential evapotranspiration
29	qtile	mm	subsurface tile flow leaving the landscape
30	irr	mm	irrigation water applied
31	surq_runon	mm	surface runoff received from upland landscape
32	latq_runon	mm	lateral flow received from upland landscape
33	overbank	mm	overbank flooding from channels
34	surq_cha	mm	surface runoff flowing into channels
35	surq_res	mm	surface runoff flowing into reservoirs
36	surq_ls	mm	surface runoff flowing onto a landscape element
37	latq_cha	mm	lateral flow into channels
38	latq_res	mm	lateral flow into reservoirs
39	latq_ls	mm	lateral flow into a landscape element
40	gwsoilq	mm	groundwater transferred to soil profile (gwflow)
41	satex	mm	saturation excess flow developed from high water table (gwflow)
42	satex_chan	mm	saturation excess flow reaching the main channel (gwflow)
43	sw_change (delsw)	mm	change in soil water volume
44	lagsurf	mm	surface runoff in transit to channel
45	laglatq	mm	lateral flow in transit to channel
46	lagsatex	mm	saturation excess flow in transit to channel
47	wet_evap	mm	evaporation from wetland surface
48	wet_out (wet_oflo)	mm	outflow (spill) from wetland
49	wet_stor	mm	wetland storage at end of timestep

The HRU writer also appends two trailing string fields after the numeric columns, written by `hru_output.f90`:

Trailing	Description
plant_cov	name of the plant community from `lum.dtl`
mgt_ops	name of the management schedule from `lum.dtl`

Aggregation

- Daily (`_day`): values for the day. Files use unit 2000 (HRU), 2050 (basin), 2140 (lsunit).
- Monthly (`_mon`): daily values summed; `sw`, `sno`, and `cn` reset their initial/final markers at month boundaries. Unit 2001 (HRU), 2051 (basin), 2141 (lsunit).
- Yearly (`_yr`): monthly values summed; storage variables divided by 12 where appropriate. Unit 2002 (HRU), 2052 (basin), 2142 (lsunit).
- Annual average (`_aa`): yearly values divided by `time%yrs_prt`. Unit 2003 (HRU), 2053 (basin), 2143 (lsunit).

Example

`refdata/Ames_sub1/basin_wb_aa.txt`:

```
demo                        SWAT+ 2024-08-27        MODULAR Rev 2024.61.0.1-33-g68b1328
 jday    mon    day     yr    unit  gis_id  name                  precip     snofall      snomlt     surq_gen
                                                                    mm          mm          mm           mm
  366     12     31   2020          1       1  demo                1905.214    220.690     219.997      59.5
```

The 42 numeric columns after `name` map one-to-one onto fields 8 to 49 above. The trailing `Original Simulation` token is the scenario name (basin scope substitutes it for the HRU `plant_cov`/`mgt_ops` pair).

Related files

- Naming convention.
- Channel and sediment outputs.
- `print.prt`. Toggles each file group on or off.